# Observer-Patch Holography Cosmological Vacuum and Structure Formation: Finite Transfer Conditions, Fluctuation Ensembles, and Proto-Objects

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**Abstract**

We specify finite quotient ensembles and transfer conditions for Observer-Patch Holography (OPH) cosmology. A declared quotient-intrinsic action and base measure determine a finite state, while reversible Euclidean data, reflection positivity and compatible refinement support its vacuum interpretation. On a supplied Whitney action, controlled real-field trajectories and a neutral interacting Hilbert space provide explicit matter and state constructions, and we state the further source and transfer conditions that connect such objects to primordial seeds, observer-facing worldlines and large-scale structure observables. The source screens of the executable observer construction test these conditions: their native density law has a reflection-positive reconstruction whose generator is the native continuum operator, and the full return reference approaches the normalized trace, so no quantum vacuum is identified. On the sourced causal order the expansion history is a record density. The capacity dictionary $\Lambda\ell_P^2 = 3\pi/N$, the de Sitter horizon entropy read backwards, returns, as diagnostics of two target-informed closure candidates, the cosmological constant $+0.62$ and $+0.38$ percent above the Planck central value, while the candidates sit $-2.64$ and $-2.81$ standard deviations below the DESI DR2 display of the same coordinate; under a Gaussian moment diagnostic the fixed-capacity equation of state $w = -1$ sits 2.6 to 3.9 standard deviations from the DESI DR2 posterior means.

**One theory and its executable hypothesis.** OPH proposes closure: one complete history is both a timeless self-referential structure and its own internal computation, and observers inside it recover and realize its specification. The hypothesis that makes this proposal concrete and testable is executable.

> *A simple, fully specified local repair and memory law on the proposed carrier architecture admits continuing observers whose compatible records and interventions have the observed* $3+1$*-dimensional quantum-field description and a common classical-gravity limit. The preferred matter mechanism is localized topological defects of native overlap transport, with rest energy derived from their excess maintenance dynamics in the same physical clock and energy dictionary.*

One source process must meet every part of this statement: records, clocks, matter, energy and responsive geometry come from the same law, state and refinement. Supplied coherence, protected codes, radial directions, fields, clocks or couplings are premises until that process realizes them. A failed specified law rejects that law and its tested interpretation; it rejects the wider proposal only when the rejected choice is proved necessary.

**The tower.** The papers form one tower over the executable model. Each layer states a hypothesis and proves finite theorems under displayed premises. A conclusion passes to the next layer only after that premise is proved on the same model process; until then it is a conditional implication.

0. **Closure and agreement.** One history reads itself into existence, and agreement on shared records defines the public world. This is the interpretation and acceptance criterion; no theorem selects a unique universe.

1. **Repair mathematics.** Local agreement repair reaches global consistency, and nonzero cycle holonomy is the residue that local repair cannot remove. Fixed-point, normal-form, frustration and finite event-algebra theorems hold on their stated domains.

2. **Carrier and repair law.** Federated twelve-port icosahedral carriers with the native C/E seam repair are the substrate. The seam rule follows from four stated requirements, the quantum record laws are classified, and the uniform native repair selects a rank-three slow-response band.

3. **Executable observer.** Observers are continuing native processes that prepare, read, retain and act on records. Finite complete-output executions, overlapping-observer correspondence, event clocks and checkpoint continuation are verified; one continuing observer with every service and resource accounted is the open join.

4. **Clocks and fields.** Modular flow of observer algebras supplies candidate local clocks, and native repair supplies a positive continuum generator of spectral dimension two on the screen. A declared radial extension gives a conditional smooth $1 + 3$ scalar description. A physical clock and energy normalization and an interacting chiral $3 + 1$ field description from native records are open joins.

5. **Gauge structure and matter.** Gauge structure is the transport of overlap data, and stable massive matter is a localized defect whose boundary transport carries a holonomy that admitted repairs cannot erase. The compact Lie type $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$ follows under complete-response premises, and fixed nonzero holonomy forces positive residual mismatch. No native defect sector, maintenance-energy law or measured spectrum is established; the native double-load candidate is transient.

6. **Gravity.** Einstein dynamics is the coarse response of the same process. The null-balance and tensor-completion implication is proved under its premises; realizing those premises on one refinement tower is open, and the static radial product fails null balance.

7. **Cosmology and tests.** The same process must fix the vacuum, dark sector, primordial response and black-hole records, and pass independent prospective tests. The cosmology papers and the fine-structure, Koide, de Sitter and Yang–Mills supplements carry conditional branches, diagnostics and retrospective comparisons at this layer.

**Completion and stopping.** The aim is a consistent and complete theory: every layer realized on one process, or a proved obstruction that removes OPH or a stated branch of it. Results that no join

needs leave the active chain with their provenance. A universal obstruction over a frozen admissible class, an unavoidable excluded prediction, or a contradiction among indispensable premises stops that class. A failed candidate, an unconstructed map or a resource stop has a narrower meaning. The ordered joins, retained foundations and stopping rules are recorded with the executable model in the public repository.[1]

**Fixed physical boundaries.** Repair counts, host runtime and entropy production are distinct from energy; the maintenance-to-energy map must be derived with one common signal speed and no per-species conversion. The modular parameter is dimensionless until a derived clock calibrates it. Stationary repair or record entropy is distinct from the fixed-volume generalized-entropy variation that the Einstein construction consumes. A massless photon makes the infinite-volume theory gapless, so sector gaps and thresholds name their state and limit; confinement precludes free massless gluons as observable particles; a quantum graviton lies outside the classical Einstein theorem.

## This paper's hypothesis and place in the tower

This paper sits at layer seven and develops the vacuum and structure-seed branch. Its hypothesis is that the cosmological vacuum is the refinement limit of a finite quotient ensemble whose reversible Euclidean data come from the source itself, that structure grows from the fluctuations of that vacuum through an expanding background, and that the vacuum term of the background is fixed by a screen capacity through $\Lambda\ell_P^2 = 3\pi/N$.

The finite quotient ensemble theorems and the matter and state constructions on a supplied spatial domain are proved on their stated domains; the flat massless vacuum control and the sector-weight freedom of restricted thermal states are exact. The vacuum interpretation, the seed bridge and the capacity reading are conditional: the conductances, potential and slab thickness, the expanding background, the physical clock and the region map behind an area reading of $N$ are premises.

For the branch to apply, the executable model must derive the Euclidean source data $(J_r, V_r, a_{t,r})$ from one native process, prepare a coherent state on it, and supply the expanding background, clock calibration and screen-to-volume transfer. The native construction constrains this join: its unread repair is a fair swap, so its density law has a reflection-positive reconstruction, but the reconstructed one-particle generator is the continuum operator $A$ rather than $\sqrt{A}$, and the full return reference approaches the normalized trace, so no quantum vacuum is identified (Section 4). The branch is refuted by a proof that no admissible source data give a reflection-positive vacuum with a relativistic one-particle generator; its fixed-capacity background is refuted if the frozen equation-of-state protocol excludes $(w_0, w_a) = (-1, 0)$ on its eligible data.

## 1 Scope and the capacity background

The executable observer construction [6, 3] gives a positive screen continuum operator of spectral dimension two [7] and does not supply the source Euclidean slab, expanding background or transfer conditions stated below. The simulator source and its public data archives are cited in [5]; the large finite repair archive is a diagnostic that passes neither the physical-CMB nor the production-gravity criterion.

[1] https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/docs/COMPLETION_PROGRAM.md

The vacuum term of the background has one dictionary on the fixed-capacity branch, $\Lambda\ell_P^2 = 3\pi/N$. Reading $N$ as de Sitter horizon entropy is a premise, and with it the dictionary is that entropy, $N = 3\pi/(\Lambda\ell_P^2)$ in nats, inverted: it holds by definition and fixes no independent value of $N$. The comparison below therefore tests candidate formulas for $N$, not the dictionary. On the native source, regional reduced entropy grows with port count and mutual information per crossing seam tends to zero, so a whole-screen area reading requires a region map that the executable construction does not supply [11]. The two closure candidates for the capacity, $3.2921 \times 10^{122}$ and $3.3001 \times 10^{122}$, return $\Lambda = 1.0959 \times 10^{-52}$ and $1.0933 \times 10^{-52}\,\mathrm{m}^{-2}$ against the Planck-central $1.0891 \times 10^{-52}\,\mathrm{m}^{-2}$, residuals of $+0.62$ and $+0.38$ percent, and sit 2.64 and 2.81 posterior standard deviations below the DESI DR2 with CMB display $(2.968 \pm 0.040) \times 10^{-122}$; both brackets are exact rational inequalities machine-checked in the Lean module `CosmologyLedgerBrackets`. Both candidates are target-informed declared maps exposed after the data, so the residual is a diagnostic of a frozen formula and carries no predictive weight; a relative capacity residual $\delta$ gives a relative constant residual $-\delta/(1+\delta)$, so the large capacity explains the small scale and not the closeness. On the same branch a constant capacity gives $w = -1$ exactly and a monotone capacity gives $w \geq -1$; the Gaussian moment diagnostic of the official DESI DR2 chains places $(-1, 0)$ between $2.6\sigma$ and $3.9\sigma$ from the four default posterior means, with monotone-branch posterior fractions between 0.007 and 0.11 percent, so the fixed branch is under tension on seen data. The conditional FZ-13 protocol was frozen on 2026-10-04 with register status `frozen_stamped_upgrade_pending`; only its complete protocol and anchoring requirements define the eligible comparison. The fixed-capacity point coincides with ΛCDM, so its survival earns no distinctive OPH support. FZ-01, frozen on 2026-07-17, registers the capacity band $N \in [3.30836, 3.31137] \times 10^{122}$ and the point $(w_0, w_a) = (-1, 0)$ without a registered verdict; both closure candidates lie below that band [12]. The cosmology postdiction ledger released with these papers records these rows with their classes.

A cosmological vacuum requires the state, expanding geometry, physical clock calibration and screen-to-volume transfer conditions below; smooth Einstein promotion of the background requires same-family tensor-curvature convergence or the continuum small-ball/null-balance identification [1].

## 2 Matter and States on a Supplied Spatial Domain

The Whitney construction on the fixed tetrahedral cone gives a controlled matter continuum in a specified sector of one charged scalar action. For $m^2 > 0$, $g \geq 0$, a supplied $u \in C^2([0,T]; H^2(\Omega))$ solving the real cubic Klein–Gordon equation with natural Neumann boundary, and Ritz-projected initial data, uniform conforming refinements give

$$\sup_{0 \leq t \leq T} \left( \|u_n - u\|_{H^1} + \|\dot{u}_n - u_t\|_{L^2} \right) \leq C_T n^{-1}.$$

The real, zero-current solutions are invariant solutions of the full charged action. The proof uses a fixed shape-regular refinement family, the conserved nonlinear energy and a Ritz error estimate, as proved in *Observers Are All You Need* [2].

On the original finite mesh with nonzero coupling $e$, eliminating mean-zero gauge directions gives a complete kinetic metric $\gamma$ on $\mathcal{Q} \cong \mathbb{R}^{30} \times \mathbb{C}^{13}$. With supplied $\hbar > 0$ and Laplace–Beltrami ordering, the neutral subspace of $L^2(\mathcal{Q}, d\mathrm{vol}_\gamma)$ has a unique nonnegative self-adjoint Hamiltonian and unitary dynamics. Normalized Gaussian half-density states lie in its operator domain and have explicit initial matter moments [2]. These are states of a finite spatial discretization, not a derived cosmological ground state or a continuum quantum field limit.

For the same autonomous action, the Jacobi duration

$$d\tau = \sqrt{\frac{G_q(dq, dq)}{2(E - V(q))}}$$

recovers action time on regular nonturning solutions at supplied energy $E$, using the full coupled kinetic metric or its gauge reduction [2]. Its configuration-based reconstruction supplies an internal model clock. A cosmological interpretation additionally requires observer-selected geometry, physical preparation and units, and a compatible expanding background. Those attachments are separate from these matter and state theorems.

The golden source population of the spacetime construction carries a second matter and state construction on the same population that the causal order uses. Its positive source-metric scalar action has a finite-time continuum limit with an explicit detector error bound, a canonical quantization whose quadratic Hamiltonian has a unique normalized ground state, the original action vacuum, and a bounded probability comparison between the split evolution of that vacuum with a coherent displacement and the continuous evolution of the same finite action, with all sixty-four modes retained at $q = 5$. The joint spatial and temporal refinement of one compact detector is certified at $q = 233$, with total error below 0.0501 against a resolved response above 0.1377. This is a vacuum state of a finite scalar action on the source population, with the population, action and tick supplied. It is neither a reflection-positive Euclidean vacuum of a cosmological release law nor a source-instantiated triple $(J_r, V_r, a_{t,r})$ in the sense of the next section, and it carries no expanding background.

On the same family a spatially flat expanding background is a record density: the scale factor enters only as the weight $a^4$ per comoving cell and conformal tick, and the redshift is the fourth root of a record-count ratio of congruent comoving windows with constant scale factor within each, or in the shrinking-window limit of a continuous profile. The finite-source cosmology paper proves these statements with their Lean module [4]. The record-production profile is a supplied datum; the vacuum and fluctuation conditions below concern the field on that background and do not select the profile.

## 3 Finite Quotient Ensemble Theorem

**Finite quotient ensemble theorem.** Fix a finite regulator $r$. The physical presentation space is $\Sigma_r$, the presentation redundancy groupoid is $\Gamma_r$, and the finite physical quotient is

$$Q_r = \Sigma_r/\Gamma_r, \qquad \pi_r : \Sigma_r \to Q_r.$$

The quotient removes nonphysical presentation data: gauge representatives, port relabelings, mesh labels, shard or worker identifiers, queue order, repair schedule identifiers, retry counters, timestamps unless declared semantic, hidden carrier coordinates, and inert ancillary labels. If only settled configurations carry probability, the probability space is the normal-form subset

$$N_r = n_r(Q_r).$$

The map $n_r$ is a normal-form map, not a probability law. Any physical branch inherits this firewall: it must declare the quotient-intrinsic source law or action before a normal form can be read as a selection or prediction claim.

**Observable algebras and reference states.** In the finite classical case the quotient observable algebra is

$$\mathcal{O}_r = \ell^\infty(Q_r).$$

In the finite quantum case the physical algebra is a declared quotient algebra $\mathcal{A}_r^{\mathrm{phys}}$ with state

$$\omega_r(A) = \mathrm{Tr}(\rho_r A).$$

When the reference object is obtained from a finite lifted carrier, the load-bearing data are not an abstract groupoid cardinality alone. They are a tracially pointed quotient

$$\left(\mathcal{A}_{r,b}^{\mathrm{phys}}, \tau_{r,b}^0\right), \qquad \mathcal{A}_{r,b}^{\mathrm{phys}} = z_{r,b} B(\widetilde{\mathcal{H}}_r)^{G_r} z_{r,b},$$

where $U_r : G_r \to U(\widetilde{\mathcal{H}}_r)$ is the compact gauge action and $z_{r,b}$ is the central projection for the declared boundary or superselection sector. The reference trace is

$$\tau_{r,b}^0(A) = \frac{\mathrm{Tr}_{\widetilde{\mathcal{H}}_r}(A)}{\mathrm{Tr}_{\widetilde{\mathcal{H}}_r}(z_{r,b})}.$$

If

$$z_{r,b}\widetilde{\mathcal{H}}_r \cong \bigoplus_\alpha V_\alpha \otimes M_\alpha,$$

with $d_\alpha = \dim V_\alpha$ and $m_\alpha = \dim M_\alpha$, then

$$\mathcal{A}_{r,b}^{\mathrm{phys}} \cong \bigoplus_\alpha I_{V_\alpha} \otimes B(M_\alpha), \qquad p_{r,\alpha} = \frac{d_\alpha m_\alpha}{\sum_\beta d_\beta m_\beta}.$$

These are the induced central-sector weights only after the carrier representation and boundary sector have been fixed.

**OPH quotient ensemble.** An OPH quotient ensemble is specified by a quotient-intrinsic base weight and action

$$m_r : Q_r \to \mathbb{R}_{>0}, \qquad S_r : Q_r \to \mathbb{R} \cup \{+\infty\},$$

and

$$w_r(q) = m_r(q) e^{-S_r(q)}, \qquad Z_r = \sum_{q \in Q_r} w_r(q), \qquad \mu_r(q) = Z_r^{-1} w_r(q).$$

At least one action value must be finite, so that $Z_r > 0$. For a law on settled configurations only, restrict this construction to $N_r$, or set $S_r = +\infty$ outside $N_r$. A separately specified prior $\nu_r$ on $Q_r$ instead induces the settled law $\mu_r^{\mathrm{nf}} = (n_r)_\# \nu_r$. These constructions agree only if

$$\sum_{x:\, n_r(x) = q} \nu_r(x) = \mu_r(q) \qquad (q \in N_r)$$

and $\mu_r$ has no mass outside $N_r$. For example, a retraction $n(0) = 0, n(1) = n(2) = 1$ sends the uniform law on three states to weights $(1/3, 2/3)$, whereas uniform counting on its two normal forms gives $(1/2, 1/2)$. Uniform quotient counting, uniform representative counting pushed to the quotient, groupoid weights, and tracial central-sector weights are different physical claims, and a physical branch declares which one it uses.

**Normal-form projector non-selection.** For any retraction $N : Q \to Q_{\mathrm{nf}}$ onto a subset $Q_{\mathrm{nf}} \subseteq Q$, that is, any map whose restriction to $Q_{\mathrm{nf}}$ is the identity, the induced map on laws

$$\mathcal{C}_Q(\mu) = N_{\#}\mu$$

is idempotent:

$$\mathcal{C}_Q^2 = \mathcal{C}_Q.$$

Every law supported on $Q_{\mathrm{nf}}$ is fixed; both statements use the retraction property. Therefore settlement or canonicalization never selects a unique physical probability law by itself.

**Selection-gap corollary.** Let $X \subseteq Q_{\mathrm{nf}}$ be a finite set of quotient-normal candidates distinguished by visible invariants. Normal-form data determine $X$ and its quotient-visible invariants, but they do not choose a member of $X$. If two laws $\mu, \nu$ are supported on $X$ and concentrate on different candidates, both are fixed by $\mathcal{C}_Q$. Unique sector selection therefore requires source data: an intrinsic action with a unique minimizer, a declared physical ensemble, or a refinement-stable gap certificate. A defect or holonomy classification can classify possible sectors without choosing the physical sector, and a contraction or repair generator can certify convergence toward a declared target without creating the target law.

**Finite MaxEnt quotient ensemble.** For finite $Q$, positive $m$, and quotient observables $F_1, \ldots, F_k$, maximizing

$$\mathcal{H}_m(\nu) = -\sum_q \nu(q) \log \frac{\nu(q)}{m(q)}$$

subject to

$$\sum_q \nu(q) = 1, \qquad \sum_q \nu(q) F_a(q) = c_a$$

has the full-support solution, when the feasible full-support surface is nonempty,

$$\mu(q) = \frac{m(q) \exp[-\sum_a \theta_a F_a(q)]}{Z(\theta)}.$$

Boundary optima obey the same formula after restricting to their support. On a finite noncommutative quotient algebra with faithful reference state $\sigma_r$, self-adjoint constraints and a feasible set containing a faithful state,

$$\rho_r = \frac{\exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}{\operatorname{Tr} \exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}.$$

If the feasible states lie in a proper support projection $P$, first restrict to their joint support. The same multiplier argument uses $P(\log \sigma_r)P$ and $PF_{r,a}P$, exponentiated and normalized on $P\mathcal{H}$; replacing the reference by $P\sigma_r P$ before taking its logarithm is not valid without a commutation condition. An empty feasible set has no selected state. The finite constraint ledger must name every $F_{r,a}$, its units and support, the target expectation and source, sector or zero-mode treatment, refinement transformation, and proof that no run output or observational output entered the source definition.

**Refinement compatibility and RG closure.** For $s \succeq r$, let $c_{sr} : Q_s \to Q_r$ be the physical coarse maps, with $c_{tr} = c_{sr} \circ c_{ts}$ for $t \succeq s \succeq r$. Exact compatibility of weighted ensembles is equivalent to the fiber-sum identity

$$\sum_{q' :\, c_{sr}(q')=q} m_s(q') e^{-S_s(q')} = \alpha_{sr} m_r(q) e^{-S_r(q)}$$

for a constant $\alpha_{sr} > 0$ independent of $q$. Then

$$(c_{sr})_{\#} \mu_s = \mu_r.$$

If the one-step defects are

$$\delta_{k+1,k} = \left\| (c_{k+1,k})_{\#} \mu_{k+1} - \mu_k \right\|_{\mathrm{TV}},$$

then

$$\left\| (c_{nr})_{\#} \mu_n - \mu_r \right\|_{\mathrm{TV}} \le \sum_{k=r}^{n-1} \delta_{k+1,k}.$$

For exponential-family refinement, exact closure requires the fine conditional free energy

$$G_{sr,\theta}(q_r) = -\log \mathbb{E}_{m_s^0} \left[ \exp[-\theta \cdot F_s(Q_s)] \mid c_{sr}(Q_s) = q_r \right]$$

to equal $\kappa_{sr}(\theta) + R_{sr}(\theta) \cdot F_r(q_r)$. If the residual is uniformly bounded by $\varepsilon$, the induced total-variation defect is bounded by $\tanh \varepsilon$.

**Implementation invariance and representative lifting.** If implementations $A, B$ have quotient bijections $h_r : Q_r^A \to Q_r^B$ satisfying

$$m_r^B(h_r q) = m_r^A(q), \qquad S_r^B(h_r q) = S_r^A(q), \qquad h_r \circ c_{sr}^A = c_{sr}^B \circ h_s,$$

then

$$(h_r)_{\#} \mu_r^A = \mu_r^B.$$

For tracially pointed quantum quotients the corresponding equivalence is a trace-preserving quotient equivalence. It is invariant under unitary intertwiners preserving the gauge action and sector, and under inert trivial ancillas $A \mapsto A \otimes I_{\mathrm{anc}}$. It is not invariant under arbitrary changes of gauge-representation multiplicities.

If an implementation stores representatives, a representative-level law must be a conditional lift

$$\widetilde{\mu}_r(x) = \mu_r(\pi_r x) \kappa_r(x \mid \pi_r x), \qquad \sum_{x : \pi_r(x) = q} \kappa_r(x \mid q) = 1.$$

Then $(\pi_r)_{\#} \widetilde{\mu}_r = \mu_r$. Uniform representative sampling yields orbit-size weights and is physical only if representative counting is the declared base measure.

**Quotient-lumpable kernels and sampler correctness.** A representative kernel $\widetilde{P}(x, y)$ descends to $Q_r$ only when

$$P_Q(q, q') = \sum_{y : \pi(y) = q'} \widetilde{P}(x, y)$$

is independent of the chosen representative $x \in \pi^{-1}(q)$. For $w(q) = m(q) e^{-S(q)}$ and proposal $R(q, q')$ with reciprocal support, the Metropolis–Hastings acceptance rule

$$a(q, q') = \min \left\{ 1, \frac{w(q') R(q', q)}{w(q) R(q, q')} \right\}$$

gives detailed balance

$$\mu(q)R(q,q')a(q,q') = \mu(q')R(q',q)a(q',q).$$

Repair-informed proposals must include the Hastings asymmetry term; otherwise the stationary law is generically changed.

**Repair generators are not selectors.** A repair generator of the form

$$L_{\mathrm{rep}} = \sum_C c_C(I - E_C)$$

is a relaxation or sampling object after a law has been selected. Conditional expectations $E_C$ are defined on $L^2(X_r, \pi_r)$, so the reference law $\pi_r$ is input. On overlapping collars the expectations need not commute. The correct finite gap certificate is the Poincare constant on the orthogonal complement of the common fixed space $\mathcal{F}_r = \bigcap_C \operatorname{Ran} E_C$:

$$\kappa_r = \inf_{0 \neq f \perp \mathcal{F}_r} \frac{\sum_C \|(I - E_C)f\|^2}{\|f\|^2}.$$

If every $c_C \geq \gamma_* > 0$, then

$$L_{\mathrm{rep}} \geq \gamma_* \kappa_r (I - P_0),$$

where $P_0$ projects onto $\mathcal{F}_r$. In finite dimension, $\kappa_r > 0$ whenever $\mathcal{F}_r^{\perp} \neq \{0\}$, since zero Dirichlet energy would put $f$ in every fixed range. If all states are fixed, the displayed operator inequality is vacuous. Replacing $f \perp \mathcal{F}_r$ by $f \perp 1$ requires the additional ergodicity condition $\mathcal{F}_r = \operatorname{span}\{1\}$; a conserved nonconstant record otherwise makes that infimum zero. A uniform refinement lower bound $\inf_r \kappa_r > 0$ is a separate theorem or receipt. Multiplying all rates by $s > 0$ rescales the generator and gap by $s$, while leaving the fixed space, stationarity and detailed balance unchanged. Numerical recognition therefore tests identities after removing a common rate scale and restores that scale to reported gaps. An unresolved positive gap is a precision limitation, not evidence that changing clock units changed the equilibrium.

**Finite evidence accuracy.** For bounded coarse observables $O$, if

$$\|\widehat{\mu}_s - \mu_s\|_{\mathrm{TV}} \leq \epsilon_{\mathrm{samp}}$$

and the refinement defects sum to $\epsilon_{\mathrm{ref}}$, then

$$\left|\mathbb{E}_{\widehat{\mu}_s}[O \circ c_{sr}] - \mathbb{E}_{\mu_r}[O]\right| \leq 2\|O\|_\infty(\epsilon_{\mathrm{samp}} + \epsilon_{\mathrm{ref}}).$$

Continuum-facing observables require a realization map and correlation Cauchy bound in addition to a finite histogram.

**Physical-vacuum requirement.** A stationary sampler is not a physical vacuum. For any faithful target law one can build a positive transfer operator with that law as ground state, so positivity alone is not a selector. A physical-vacuum interpretation requires source Euclidean slab data

$$\mathfrak{S}_r^E = (Q_r, m_r^0, J_r, V_r, a_{t,r})$$

whose conductance $J_r(q,q') = J_r(q',q) \geq 0$, local potential $V_r$, and slab thickness $a_{t,r}$ are derived without using the target law or sampler output. With connected event graph,

$$(H_r^E f)(q) = \frac{1}{m_r^0(q)} \sum_{q'} J_r(q,q')(f(q) - f(q')) + V_r(q)f(q)$$

is self-adjoint and bounded below on $L^2(Q_r, m_r^0)$; its finite Feynman–Kac semigroup is positivity improving. Perron–Frobenius gives a unique positive normalized ground state $\Omega_r$, and the finite vacuum law is

$$\mu_r^{\mathrm{vac}}(q) = |\Omega_r(q)|^2 m_r^0(q).$$

For $T_r = e^{-a_{t,r}(H_r^E - E_{0,r})}$, the Doob kernel is stochastic and detailed-balanced with $\mu_r^{\mathrm{vac}}$. A continuum interpretation additionally requires reflection positivity or equivalent reconstruction plus refinement compatibility of the transfer family. On the native source screens the actual density law has such a reflection-positive reconstruction along one common refining subsequence; its one-particle generator is the native continuum operator $A$ rather than $\sqrt{A}$, and it supplies no coherent preparation, relativistic vacuum or Lorentz symmetry.[2]

**Primordial and cosmological prediction firewall.** A screen covariance contains incomplete radial information. The complete one-shell map

$$C_\ell = 4\pi \int_0^\infty \frac{dk}{k} \Delta_\zeta^2(k) j_\ell^2(k\chi_\star)$$

has an infinite-dimensional kernel that persists under positivity. OPH physical primordial interpretation requires the source-only stress, single-clock, entropy-repair, curvature-evolution, adiabatic-mode, phase-coherence, physical mode, radial-null-space, and forward-projection receipts together with a scale-natural physical dilation intertwiner or complete radial cross-covariance tomography. A finite radial prior produces a conditional continuation. Observable CMB comparison also requires declared source, solver, dataset, covariance, nuisance, data-use, and pooled-reducer provenance. On the native source screens the continuum energy is not boundedly congruent to the static boundary action of a smooth compact three-dimensional Dirichlet bulk, whose Steklov eigenvalues grow like $\sqrt{n}$ while the native eigenvalues grow like $n$, and internal rank-three carrier modes vanish from bounded-energy families; a radial lift needs a radial observation algebra that the native energy does not supply.[3]

**Native source screens.** On the native source screens the actual density law has the reflection-positive reconstruction that a continuum interpretation requires, along one common refining subsequence; it supplies no coherent preparation, relativistic vacuum or Lorentz symmetry [8] (Section 4). The same continuum energy is not boundedly congruent to the static boundary action of a smooth compact three-dimensional Dirichlet bulk, whose Steklov eigenvalues grow like $\sqrt{n}$ while the native eigenvalues grow like $n$, and internal rank-three carrier modes vanish from bounded-energy families; a radial lift needs a radial observation algebra that the native energy does not supply [7].

## 4 Vacuum Claim Boundary

A simulated field is classified as seed noise, repair jitter, or a conventional reference ensemble unless it satisfies the finite quotient ensemble conditions. An OPH-native vacuum requires source data of the form

$$\mathfrak{S}_r^E = (Q_r, m_r^0, J_r, V_r, a_{t,r})$$

[2] https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/observer_time_fields/REFINEMENT.md

[3] https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/physical_space_time/OCCUPATION_FIELD.md

with $J_r$, $V_r$, and $a_{t,r}$ derived from OPH source structure rather than from a target law or a desired visual spectrum. Directed accepted repair moves cannot be reused as reversible Euclidean conductances. The native seam rule differs: each unread repair has the law of a fair identity-or-endpoint swap, so its configuration transfer $K$ is symmetric and positive, and the actual native density law has a reflection-positive Gaussian reconstruction. The obstruction on the native source lies elsewhere: the reconstructed one-particle generator is the continuum operator $A$, not $\sqrt{A}$, and the full configuration return reference built from $K$ approaches the normalized trace at fixed modular parameter and gives the density-wave sector vanishing weight, so it identifies no quantum vacuum [8].

### 4.1 Vacuum and initial-state readouts on the effective product

The active product construction has a different, conditional state contract. With its positive spatial operator $B$, canonical fields and supplied inverse temperature $b$, its ordered free-field two-point function is

$$C_b(t) = \frac{\coth(b\sqrt{B}/2)\cos(t\sqrt{B}) - i\sin(t\sqrt{B})}{2\sqrt{B}}.$$

The limit $b \to \infty$ gives the static ground-state covariance $(2\sqrt{B})^{-1}$; finite $b$ changes fluctuations while preserving the same commutator and response. These are joint observer-readout predictions of the declared free-field family, not the passive native density law or the interacting field's covariance [8]. The radial apparatus, state preparation and physical clock map remain inputs.

On the finite interacting work algebra $\mathcal{A} = M_8(\mathbb{C}) \oplus M_8(\mathbb{C})$, restricted thermal equilibrium leaves a central sector weight:

$$\rho_w = w\frac{e^{-bH_+}}{Z_+} \oplus (1-w)\frac{e^{-bH_-}}{Z_-}, \qquad 0 < w < 1.$$

Every such state has the same restricted modular flow. Cooling each block does not select $w$. The full Gibbs preparation additionally chooses $w = Z_+/(Z_+ + Z_-)$; its global ground-state limit is therefore a stronger state choice. Admitted record statistics can test sector populations, but do not identify an unobserved purification [9].

A direct cosmological identification has a discriminating control. In the flat massless limit, with $\hbar = c = 1$ and $\langle \phi_{\mathbf{k}} \phi_{\mathbf{k}'} \rangle = (2\pi)^3 \delta(\mathbf{k} + \mathbf{k}') P_\phi(k)$, the static vacuum gives

$$P_\phi(k) = \frac{1}{2k}, \qquad \Delta^2_\phi(k) = \frac{k^3 P_\phi(k)}{2\pi^2} = \frac{k^2}{4\pi^2}.$$

Thus a scale-independent identification of this field with primordial curvature does not reproduce the nearly scale-invariant curvature spectrum inferred from CMB data [10]. This flat-limit control is not a Fourier-spectrum claim for the bounded native radial product. An expanding evolution and an operational curvature map must supply the transfer before the state can predict cosmological observations. The initial-state target is consequently a law for accessible earlier-state correlations and their later records, not an assumed observable absolute beginning of the Universe.

## 5 Structure-Formation Requirements

| Object | Required source-side label |
|---|---|
| Fluctuating screen field | Seed noise or a conventional reference ensemble unless the quotient ensemble conditions hold. |
| OPH vacuum field | Source Euclidean slab, ground-state, Doob, reflection-positivity, and transfer-refinement conditions. |
| Proto-object worldlines | Diagnostic until production matter/particle and observer-facing object population conditions hold. |
| Growth seeds | Primordial bridge, physical mode basis, radial-null, positivity, forward-projection, and source-dilation or tomography conditions. |
| Large-scale structure comparison | Diagnostic; source, transfer, and official likelihood/data contracts are not supplied. |

## 6 Numerical Acceptance Conditions

The finite-patch calculation reports the following five conditions separately:

- a tracially pointed quotient;
- a source-derived Euclidean slab;
- transfer stability under refinement;
- schedule invariance of the stationary law;
- pathwise invariance under partition changes.

A failed condition is part of the scientific result and excludes the physical vacuum classification for a reference baseline or rendering field.

## 7 Unsupplied Premises

The cosmological source construction does not supply the reversible conductances $J_r$, potential $V_r$, or slab thickness $a_{t,r}$ required above. A source-instantiated reflection-positive vacuum, or an equivalent reconstruction on this cosmological branch, is not supplied. The bridge from a source-derived vacuum to primordial growth seeds without cosmic microwave background or large-scale structure data is not supplied. A physical proto-object or worldline requires conditions distinct from those used for production particle claims.

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